```
> restart;#PSTc
> ρ := 9.9·10^-37; TQ := 1.765·10^-19; SQ := 4.25·10^-19; c := 299792458; Rs :=
    1.553479855 × 10^26; R := 3.922833072 × 10^22;# ρ  density of void space;
```

$$\rho := 9.900000000 \times 10^{-37}$$

$$TQ := 1.765000000 \times 10^{-19}$$

$$SQ := 4.250000000 \times 10^{-19}$$

$$c := 299792458$$

$$Rs := 1.553479855 \times 10^{26}$$

$$R := 3.922833072 \times 10^{22} \quad (1)$$

```
> V := χ^0.5 / ρ^0.5;
```

$$V := 1.005037815 \times 10^{18} \sqrt{\chi} \quad (2)$$

```
> χ := 0.09070366748; V;
```

$$\chi := 0.09070366748$$

$$3.026877369 \times 10^{17} \quad (3)$$

```
> 4·Pi/3 (Rs)^3 / SQ^3;# approx. number of space quantum bloks in LU
```

$$2.045684699 \times 10^{134} \quad (4)$$

```
> T = 2·Rs / V; PTU := 3·24·3600;# 3 days is the unit
```

$$T = 1.026457081 \times 10^{9}$$

$$PTU := 259200 \quad (5)$$

```
> PTU := 3·24·3600; 2·R / V;#The PTU is 3 days is the unit;
```

$$PTU := 259200$$

$$259200.0002 \quad (6)$$

```
> PTQ := evalf(SQ / c);
```

$$PTQ := 1.417647405 \times 10^{-27} \quad (7)$$

```
> V := 2·R / 3·24·3600; PTU := 3·24·3600; 2·R / V;# 3 days is the unit
```

$$V := 3.026877370 \times 10^{17}$$

$$PTU := 259200$$

$$259200.0000 \quad (8)$$

```
> Vp := 2·R / PTU# The speed limit of a wave
```

$$Vp := 3.026877370 \times 10^{17} \quad (9)$$

```
> Vp / 24·3·c;# The speed limit of un object with null rest mass~14
    ·10^6times the speed of light 14·10^6 LS
```

$$1.402302238 \times 10^{7} \quad (10)$$

```
> χ := (Vp·ρ^0.5)^2;
```

(11)

$$\chi := 0.09070366748 \tag{11}$$

```
> 4·PTQ;
```

$$5.670589620 \times 10^{-27} \tag{12}$$

```
> 1/R;# is the rough aproximation of photon radius in [m]
```

$$2.549178060 \times 10^{-23} \tag{13}$$

```
> rb := 5.292410932 × 10^-11; rb^2;
```

$$rb := 5.292410932 \times 10^{-11}$$

$$2.800961347 \times 10^{-21} \tag{14}$$

```
> Sq := 4.3·10^-19; Sq^2;
```

$$Sq := 4.300000000 \times 10^{-19}$$

$$1.849000000 \times 10^{-37} \tag{15}$$

```
> SQ^2;# The Quark dimmension 4.25-4.3·10^-19
```

$$1.806250000 \times 10^{-37} \tag{16}$$

```
> PTQ^2
```

$$2.009724165 \times 10^{-54} \tag{17}$$

```
> Fτ := -9.897187240 × 10^23/r^2 + 3.121665052 × 10^21/r - 0.07957747155;
```

$$F\tau := -\frac{9.897187240 \times 10^{23}}{r^2} + \frac{3.121665052 \times 10^{21}}{r} - 0.07957747155 \tag{18}$$

```
> simplify(-.9914018010e24/r^2+.3121665052e22/r-.7957747152e-1);
```

$$\frac{-9.914018010 \times 10^{23} + 3.121665052 \times 10^{21}\, r - 0.07957747152\, r^2}{r^2} \tag{19}$$

```
>
> restart;# The Gamma Ray spherical simetry;
>  de := diff(g(r), r$2) = r·g(r)/Rs;
```

$$de := \frac{d^2}{dr^2}\, g(r) = \frac{r\, g(r)}{Rs} \tag{20}$$

```
> ics := g( 2·Rs) = 0 , D(g)(0) = R;
```

$$ics := g(2\, Rs) = 0,\ D(g)(0) = R \tag{21}$$

```
> sol := dsolve({de, ics});
```

$$sol := g(r)$$

$$= \frac{2\, R\, \pi\, \mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right) \mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3} r\right)}{\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right) + 3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right)\right)}$$

$$- \left(2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right) R\, \pi\, \mathrm{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3} r\right)\right) \Big/ \left(\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(\right.\right.\right.$$

$$\left.\left.\left.{}^{3}\, Rs\right)\right)\right)$$

```
>  g1(r) := rhs(sol);
```

$$g1 := r \mapsto rhs(sol) \tag{23}$$

```
> g1(0);
```

$$\frac{2\,R\,\pi\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}r\right)}{\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)}$$

$$-\left(2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)R\,\pi\,\mathrm{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}r\right)\right)\Big/\left(\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\mathrm{AiryAi}\left(\right.\right.\right.$$

$$\left.\left.\left.{}^{3}Rs\right)\right)\right)$$

```
> g(r) := (2 R π AiryBi(-2 (-1/Rs)^(1/3) Rs) AiryAi(-(-1/Rs)^(1/3) r))/(Γ(2/3) (-1/Rs)^(1/
      3) (3^(2/3) AiryAi(-2 (-1/Rs)^(1/3) Rs) + 3^(1/6) AiryBi(-2 (-1/Rs)^(1/3) Rs)))
      - (2 AiryAi(-2 (-1/Rs)^(1/3) Rs) R π AiryBi(-(-1/Rs)^(1/3) r))/(Γ(2/3) (-1/Rs)^(1/3) (3
      Rs)));
```

$$g := r \mapsto \left(2\cdot R\cdot\pi\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot\mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}\cdot r\right)\right)\Big/\left(\Gamma\left(\frac{2}{3}\right)\cdot\left(\right.\right.$$

$$\left.\left.-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)+3^{1/6}\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\right)\right)$$

$$-\left(2\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot R\cdot\pi\cdot\mathrm{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}\cdot r\right)\right)\Big/\left(\Gamma\left(\frac{2}{3}\right)\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{Airy}\right.\right.$$

$$\left.\left.\cdot Rs\right)\right)\Big)$$

```
> # g(r)·∂u(t)/∂t + u(t)· ∂g(r)/∂r·∂r/∂t = c ;
    # thus for r = 0 and t≠0 we have u(t) adimensional;
> eq1 := (g(0)·diff(u(t), t) + u(t)·R·V - V = 0);
```

$eq1 :=$ **(26)**

$$\left(\left(2\,R\,\pi\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)\Big/\left(3\,\Gamma\left(\frac{2}{3}\right)^{2}\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\mathrm{AiryAi}\left(\right.\right.\right.\right.$$

$$\left.\left.\left.\left.-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)\right)$$

$$-\left(2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)R\,\pi\,3^{5/6}\right)\Big/\left(3\,\Gamma\left(\frac{2}{3}\right)^{2}\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\mathrm{AiryAi}\left(\right.\right.\right.$$

$$\left.\left.\left.-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)\right)\left(\frac{\mathrm{d}}{\mathrm{d}t}u(t)\right)+u(t)\,R\,V-V=0$$

```
> dsolve(eq1, u(t));
```

$$u(t) = \left(\frac{e^{-\frac{3\,V\Gamma\left(\frac{2}{3}\right)^2\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)t}{2\pi\left(\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}-\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}}}{R} + _C1 \right) e^{-\frac{3\,V\Gamma\left(\frac{2}{3}\right)^2\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)t}{2\pi\left(-\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}+\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}} \tag{27}$$

> $u(t) := \left(\frac{e^{-\frac{3\,V\,\Gamma\left(\frac{2}{3}\right)^2\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)t}{2\,\pi\left(\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}-\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}}}{R} + _C1 \right) e^{-\frac{3\,V\,\Gamma\left(\frac{2}{3}\right)^2\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)t}{2\,\pi\left(-\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}+\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}}$

$$u := t \mapsto \left(\frac{e^{-\frac{3\cdot V\cdot\Gamma\left(\frac{2}{3}\right)^2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)+3^{1/6}\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\right)\cdot t}{2\cdot\pi\cdot\left(\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{5/6}-\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{1/3}\right)}}}{R} + _C1 \right) \cdot e^{-\frac{3\cdot V\cdot\Gamma\left(\frac{2}{3}\right)^2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)+3^{1/6}\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\right)\cdot t}{2\cdot\pi\cdot\left(-\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{5/6}+\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{1/3}\right)}} \tag{28}$$

> **sol1 := u(t)·g(r); # H(r, t) = u(t)·g(r)**

$$sol1 := \left(\frac{e^{-\frac{3\,V\Gamma\left(\frac{2}{3}\right)^2\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)t}{2\pi\left(\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}-\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}}}{R} + _C1 \right)$$

$$e^{-\frac{3\,V\Gamma\left(\frac{2}{3}\right)^2\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)t}{2\pi\left(-\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}+\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}}\left(\left(2\,R\,\pi\right.\right.$$

$$\left.\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}r\right)\right)\Big/\left(\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\right.\right.$$

$$\left.\left.\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)\right)$$

$$-\left(2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)R\,\pi\,\mathrm{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}r\right)\right)\Big/\left(\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\right.\right.$$

$$\left.\left.\left.

$$\left(\left(\left(Rs^{3}\right)\right)\right)$$

>

>